# AWS Static Website Deployment

## Step-by-Step Console Guide

Module 1: Fundamentals & Setup

Module 2: Web Application Deployment

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Amazon S3 | CloudFront | Route 53 | ACM | IAM

*AWS Management Console UI Guide*

# MODULE 1 - Fundamentals & Setup

Objective: Understand basic cloud concepts and set up your AWS environment, create an S3 bucket, upload a static website, and configure public access.

## Step 1: Create an AWS Account

1. Open your browser and go to: https://aws.amazon.com
2. Click the "Create an AWS Account" button (top-right corner).
3. Enter your Email address and choose an AWS account name. Click "Verify email address".
4. Enter the verification code sent to your email. Click "Verify".
5. Create a strong Root user password and confirm it. Click "Continue".
6. On the Contact Information page, select "Personal" account type. Fill in your full name, phone number, country, and address. Click "Continue".
7. Enter your Credit/Debit card details (AWS uses this for identity verification; you will NOT be charged for Free Tier usage). Click "Verify and Continue".
8. Select the SMS or Voice call verification method. Enter your phone number and click "Send SMS".
9. Enter the 4-digit PIN received. Click "Continue".
10. Choose the "Basic support - Free" plan. Click "Complete sign up".
11. Click "Go to the AWS Management Console". Sign in with your Root email and password.

*NOTE: Always enable Multi-Factor Authentication (MFA) on your root account. Go to IAM > Security credentials > Assign MFA device.*

## Step 2: Learn Basic Cloud Concepts

### IaaS - Infrastructure as a Service

Provides virtualized computing resources over the internet. You manage OS, middleware, and applications. AWS Example: EC2 (virtual servers), S3 (storage), VPC (networking).

### PaaS - Platform as a Service

Provides a platform allowing customers to develop, run, and manage applications without managing infrastructure. AWS Example: AWS Elastic Beanstalk, AWS Lambda, AWS RDS.

### SaaS - Software as a Service

Delivers software applications over the internet on a subscription basis. The provider manages everything. AWS Example: Amazon WorkMail, Amazon Chime, AWS Marketplace SaaS apps.

**TIP: Remember: IaaS = You manage most things. PaaS = Provider manages infrastructure. SaaS = Provider manages everything.**

## Step 3: Create an Amazon S3 Bucket

1. Sign in to the AWS Management Console at https://console.aws.amazon.com
2. In the search bar at the top, type "S3" and click on "S3" under Services.
3. Click the orange "Create bucket" button.
4. Under "General configuration":
   - Bucket name: Enter a unique name e.g., "my-static-website-2024"
   - AWS Region: Select your preferred region e.g., "us-east-1 (US East N. Virginia)"
5. Under "Object Ownership":
   - Select "ACLs disabled (recommended)"
6. Under "Block Public Access settings for this bucket":
   - UNCHECK "Block all public access"
   - Check the acknowledgment checkbox that appears.
7. Leave all other settings as default.
8. Scroll down and click "Create bucket".
9. Your bucket will now appear in the S3 bucket list. Click on it to open it.

*NOTE: Bucket names must be globally unique across all AWS accounts, 3-63 characters, lowercase only, no spaces.*

## Step 4: Upload Static Website Files (HTML, CSS, JS)

First, prepare your website files locally. Create the following sample files:

**Sample index.html:**

```
<!DOCTYPE html>
<html>
  <head>
    <title>My AWS Website</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h1>Hello from AWS S3!</h1>
    <p>My static website is live on Amazon S3.</p>
    <script src="script.js"></script>
  </body>
</html>
```

1. In the AWS Console, go to S3 and click on your bucket name.
2. Click the "Upload" button.
3. Click "Add files" and select your index.html, style.css, and script.js files from your computer.
4. Alternatively, click "Add folder" to upload an entire website folder.
5. Review the files listed under "Files and folders".
6. Leave all permissions and properties as default.
7. Click the orange "Upload" button at the bottom.
8. Wait for the upload to complete. You will see a green "Upload succeeded" banner.
9. Click "Close" to return to your bucket.

## Step 5: Enable Static Website Hosting in S3

1. Inside your S3 bucket, click on the "Properties" tab.
2. Scroll all the way down to the "Static website hosting" section.
3. Click "Edit".
4. Select "Enable" under Static website hosting.
5. Hosting type: Select "Host a static website".
6. Index document: Type "index.html"
7. Error document: Type "error.html" (optional, create this file too).
8. Click "Save changes".
9. Scroll back down to "Static website hosting". You will see a Bucket website endpoint URL like:

   http://my-static-website-2024.s3-website-us-east-1.amazonaws.com

   Copy this URL - you will need it later.

*NOTE: This URL uses HTTP only. HTTPS will be enabled via CloudFront in Module 2.*

## Step 6: Configure Bucket Policy for Public Access

1. Inside your S3 bucket, click the "Permissions" tab.
2. Scroll down to "Bucket policy" and click "Edit".
3. In the Policy editor, paste the following JSON policy (replace YOUR-BUCKET-NAME):

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
```

```
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::YOUR-BUCKET-NAME/*"
    }
  ]
}
```

4. Replace "YOUR-BUCKET-NAME" with your actual bucket name.
5. Click "Save changes".
6. Go to the "Properties" tab, scroll to "Static website hosting", and click the endpoint URL to test your website in a browser.

**TIP: You should now see your HTML page loading in the browser. Module 1 is complete!**

# MODULE 2 - Web Application Deployment

Objective: Deploy and make the static website accessible globally using CloudFront CDN, Route 53 DNS, and HTTPS via AWS Certificate Manager.

## Step 7: Create CloudFront Distribution with S3 as Origin

1. In the AWS Console search bar, type "CloudFront" and click on it.
2. Click "Create a CloudFront distribution".
3. Under "Origin":
   - Origin domain: Click the dropdown and select your S3 bucket website endpoint (choose the .s3-website-... URL, NOT the .s3.amazonaws.com one)
   - Origin name: Auto-filled (leave as is)
4. Under "Default cache behavior":
   - Viewer protocol policy: Select "Redirect HTTP to HTTPS"
   - Allowed HTTP methods: Select "GET, HEAD"
   - Cache policy: Select "CachingOptimized"
5. Under "Settings":
   - Price class: Select "Use only North America and Europe" (cheapest) or "Use all edge locations"
   - Alternate domain name (CNAME): Leave empty for now (add after Route 53 setup)
   - Default root object: Type "index.html"
6. Click "Create distribution".
7. Wait 5-15 minutes for the distribution to deploy. Status changes from "Deploying" to "Enabled".
8. Copy the "Distribution domain name" (e.g., d1234abcd.cloudfront.net). Test it in your browser.

*NOTE: CloudFront caches your content at 400+ edge locations globally, making your website load faster worldwide.*

## Step 8: Register or Use a Domain Name

**Option A - Register a new domain via Route 53 (Paid ~$12/year for .com):**

1. In AWS Console, search "Route 53" and click on it.
2. Click "Register domain" in the left sidebar.
3. Type your desired domain name (e.g., mywebsite.com) and click "Search".
4. If available, click "Select" and then "Proceed to checkout".
5. Fill in your contact details (name, email, address, phone).
6. Choose auto-renewal preference and click "Submit order".

7. Check your email for a verification link from AWS. Click it to verify.
8. Domain registration takes up to 3 days but usually completes within 15-30 minutes.

**Option B - Use an existing domain from another registrar (Free):**

1. Log in to your domain registrar (GoDaddy, Namecheap, Google Domains, etc.).
2. You will update the Nameservers later after creating a Route 53 hosted zone.

**TIP: For students/testing: Use freenom.com for free .tk or .ml domains, or use the CloudFront URL directly.**

## Step 9: Configure Route 53 Hosted Zone

1. In the AWS Console, search "Route 53" and click on it.
2. In the left sidebar, click "Hosted zones".
3. Click "Create hosted zone".
4. Domain name: Enter your domain (e.g., mywebsite.com) - without "www".
5. Type: Select "Public hosted zone".
6. Click "Create hosted zone".
7. You will see 2 default records created:
   - NS (Name Server) record - contains 4 nameservers
   - SOA (Start of Authority) record
8. Copy all 4 NS (nameserver) values. Example:
   ns-123.awsdns-45.com
   ns-678.awsdns-90.org
   ns-111.awsdns-22.net
   ns-999.awsdns-55.co.uk
9. If using an external registrar: Log in to your registrar, find DNS/Nameserver settings, and replace the existing nameservers with these 4 AWS nameservers. Save changes. (DNS propagation takes up to 48 hours)

*NOTE: If you registered the domain via Route 53 in Task 8, the nameservers are automatically configured. Skip step 9.*

## Step 10: Create DNS Record Pointing to CloudFront

1. In Route 53, click on your hosted zone name to open it.
2. Click "Create record".
3. Record name: Leave empty for root domain (mywebsite.com) OR type "www" for www.mywebsite.com.
4. Record type: Select "A - Routes traffic to an IPv4 address and some AWS resources".

5. Toggle ON "Alias" (the toggle switch on the right side).
6. Route traffic to:
   - Select "Alias to CloudFront distribution"
   - Region: "US East (N. Virginia)" (CloudFront is always us-east-1)
   - Choose your CloudFront distribution from the dropdown
7. Routing policy: Select "Simple routing".
8. Click "Create records".
9. Repeat steps 2-8 to create a second record for "www" (if you did root in step 3, now do "www" and vice versa).

**TIP: Create both records - one for mywebsite.com and one for www.mywebsite.com - for complete coverage.**

## Step 11: Enable HTTPS using AWS Certificate Manager (ACM)

*NOTE: IMPORTANT: ACM certificates for CloudFront MUST be created in us-east-1 (N. Virginia) region only.*

1. In the AWS Console top-right, change your region to "US East (N. Virginia) us-east-1".
2. Search "Certificate Manager" and click on it.
3. Click "Request a certificate".
4. Select "Request a public certificate" and click "Next".
5. Fully qualified domain names: Enter both:
   - mywebsite.com

   Click "Add another name to this certificate" and add:

   - *.mywebsite.com  (covers all subdomains including www)
6. Validation method: Select "DNS validation - Recommended".
7. Key algorithm: Leave as "RSA 2048".
8. Click "Request".
9. On the certificate details page, click "Create records in Route 53" button.

   AWS will automatically add the required CNAME validation records to Route 53.
10. Wait 2-5 minutes. The Status will change from "Pending validation" to "Issued".
11. Now go to CloudFront > Click your distribution > Click "Edit".
12. Under "Alternate domain names (CNAMEs)": Add mywebsite.com and www.mywebsite.com.
13. Under "Custom SSL certificate": Select your newly issued ACM certificate from the dropdown.
14. Click "Save changes".
15. Wait 5-10 minutes for CloudFront to redeploy with the new certificate.

## Step 12: Test Website Access Using the Domain

1. Open your browser and navigate to: https://mywebsite.com
2. Verify you see a padlock icon (HTTPS) in the browser address bar.
3. Also test: https://www.mywebsite.com
4. Test HTTP redirect: Go to http://mywebsite.com - it should automatically redirect to https://
5. Open browser DevTools (F12) > Network tab > Reload the page. Check that:
   - Server header shows "CloudFront"
   - X-Cache header shows "Hit from cloudfront" (after first load)
6. Test from different locations using: https://www.whatismyip.com/website-speed-test/

**TIP: Use https://dnschecker.org to verify your DNS records have propagated globally across all regions.**

# Architecture Overview & Flow

The complete request flow for a user accessing your website:

| | |
|---|---|
| **User Browser** | User types https://mywebsite.com in browser |
| **v** | |
| **Route 53** | Resolves DNS: mywebsite.com -> CloudFront distribution IP |
| **v** | |
| **CloudFront** | Checks edge cache. If content cached (Hit) -> serves instantly.<br>If not cached (Miss) -> fetches from S3 origin. |
| **v** | |
| **Amazon S3** | Stores and serves static files (HTML, CSS, JS, images) as origin. |
| **v** | |
| **ACM / HTTPS** | SSL/TLS certificate ensures encrypted HTTPS connection end-to-end. |
| **v** | |
| **User Browser** | Receives content. Page loads fast from nearest edge location. |

## AWS Services Used - Quick Reference

| AWS Service | Purpose | Free Tier |
|---|---|---|
| Amazon S3 | Stores static website files (HTML, CSS, JS) | 5 GB storage |
| CloudFront | Global CDN - serves content from edge locations | 1 TB data/month |
| Route 53 | DNS management and domain routing | $0.50/hosted zone |
| ACM | Free SSL/TLS certificate for HTTPS | Free (public certs) |
| IAM | Identity and access management for AWS services | Always Free |

*NOTE: All tasks above use AWS Free Tier. You will NOT be charged if staying within free tier limits. Monitor your usage in the AWS Billing Dashboard.*

## Common Errors & Fixes

**ERROR: 403 Forbidden on S3 URL**

FIX: Bucket policy not set correctly OR 'Block Public Access' is still ON.

Fix: Go to S3 > Permissions > Uncheck 'Block all public access', then re-add bucket policy.

**ERROR: CloudFront shows old content**

FIX: CloudFront cached the old version.

Fix: Go to CloudFront > Invalidations > Create invalidation > Enter '/*' > Invalidate.

**ERROR: Certificate stuck in 'Pending validation'**

FIX: CNAME validation record not added to Route 53.

Fix: In ACM, click 'Create records in Route 53' button. Wait 5 minutes.

**ERROR: Website not loading on custom domain**

FIX: DNS not propagated yet OR wrong nameservers at registrar.

Fix: Wait up to 48 hours. Check propagation at dnschecker.org.

**ERROR: ACM certificate not showing in CloudFront**

FIX: Certificate was created in wrong AWS region.

Fix: Delete and recreate the certificate in us-east-1 (N. Virginia) ONLY.